UI / UX SPECIFICATION · V1.0.0 · 8 AUGUST 2026

Invest-System — interface & experience design

Invest-System is a statistical laboratory that happens to read market data — not a trading terminal. That single decision drives every interface choice in this document. A trading terminal is optimised for speed of action; this is optimised for *not being misled*. Where the two conflict, the interface slows the user down on purpose.

1 · WHO IT IS FOR, AND WHAT IT MUST PREVENT

PRIMARY USER

The researcher

Someone testing whether a relationship between two securities is real. Comfortable with a regression table; not necessarily with the traps around it.

THE FAILURE TO DESIGN AGAINST

Confident wrongness

A model that looks convincing in-sample and is worthless out of it. The interface's job is to make that visible before it is believed.

The governing principle: no number appears without its caveat attached

Most analytics tools present a coefficient and let the user infer significance, stability and generalisation. This interface treats those as part of the number. A beta always ships with its robust standard error; a classifier score always ships with its out-of-sample counterpart and the baseline it must beat; a fit that failed to converge is labelled as such and never silently rendered as a result.

2 · INFORMATION ARCHITECTURE

Five tabs, ordered as the actual research workflow runs: get data → fit a model → interrogate the estimator → get it explained → keep what matters. The order is the teaching device. A user who moves left to right is doing the work in the correct sequence without being told.

TAB	THE QUESTION IT ANSWERS	PRIMARY ACTION	STATE IT LEAVES BEHIND
Market data	What is this security worth, and how do I know?	Fetch & save	A row in the database
Model lab	Does the relationship hold, and how strongly?	Fit model	A coefficient table + diagnostics
Estimator comparison	Does the answer depend on how I solved it?	Run comparison	Convergence verdicts
AI explainer	What does this actually mean in words?	Explain results	Nothing — stateless by design
Watchlist	What am I tracking?	Add / export	A persisted list

The sidebar is a permanent honesty panel, not navigation

It never changes with the tab. It shows the data provider in use, the database schema version, an integrity flag and live row counts. The user can always answer *"where did this come from and did it save?"* without leaving what they are doing. Persistent trust indicators beat modal confirmations, which get dismissed unread.

3 · SCREEN ONE — MARKET DATA

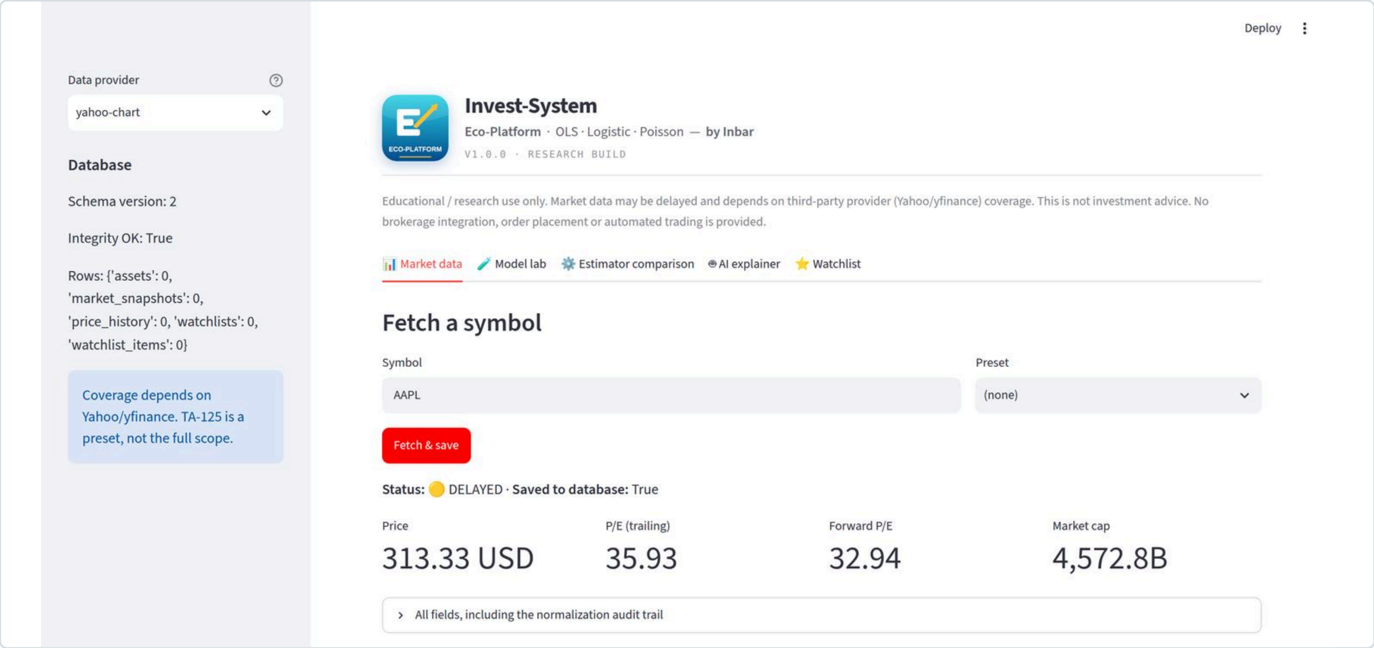


FIGURE 1 — live fetch of AAPL. Provenance badge, four valuation metrics, and a collapsed audit trail. Captured from the running application.

INTERACTION FLOW

1. **Type a symbol, or choose a preset**
   Free text accepts any provider-resolvable ticker. Presets (TA-125, multi-asset) exist so a new user never faces an empty field with no idea what is valid.

2. **One primary action**
   A single accented button. Fetching and saving are deliberately not separable — splitting them creates a state where the user believes data is stored and it is not.

3. **Provenance appears before any number**
   The status line reads `DELAYED · Saved to database: True` and sits *above* the metrics, so the caveat is read first.

4. **Four metrics, no more**
   Price, trailing P/E, forward P/E, market cap. Everything else is one click away in the audit expander rather than competing for attention.

DESIGN DECISIONS WORTH DEFENDING

Absence is explained, not left blank

Request an index or an ETF and the P/E fields read `n/a` — and a note appears stating that the asset type has no earnings, so the ratio is undefined, and that this is correct behaviour rather than a failed fetch. A blank cell makes the user doubt the tool; an explained blank teaches them something about the asset class.

Six provenance states, colour plus word

`LIVE · DELAYED · CACHED · SYNTHETIC · PARTIAL · ERROR` — each carries a coloured dot *and* the word. Colour alone would fail a colour-blind user and disappear in print. `SYNTHETIC` matters most: the app ships with an offline generator so it works with no internet, and a user must never mistake demonstration data for market data.

The audit trail is collapsed, not omitted

Raw price, raw currency, the normalization rule applied, EPS, net income and the fetch timestamp live behind one expander. Tel Aviv equities quoted in agorot are divided by 100 — and the rule that did it is recorded, so the arithmetic can be checked rather than trusted. Progressive disclosure: default calm, full detail on demand.

4 · SCREEN TWO — MODEL LAB

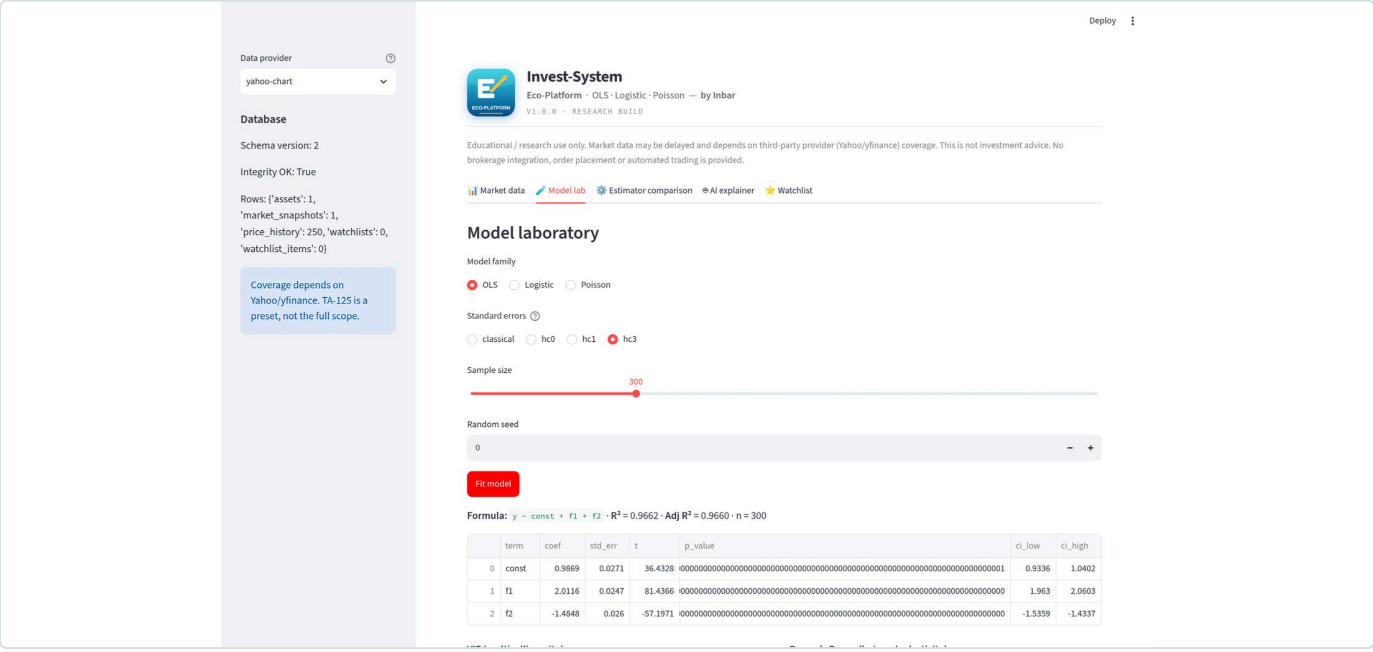


FIGURE 2 — OLS fitted on synthetic data with known parameters. Coefficient table, VIF and Breusch-Pagan panels, and the true values printed beneath so the user can verify the estimator itself.

WHY THE DEFAULTS ARE THE DESIGN

Three model families sit on one horizontal radio — OLS, Logistic, Poisson — because they are alternatives for the same task, not a hierarchy. Choosing OLS reveals a second radio for standard errors, defaulting to `HC3`.

The HC3 default is a UX decision with statistical consequences

Financial returns are almost always heteroskedastic. On real data in this application the robust standard error came out **46% larger** than the classical one — meaning the classical default, which nearly every tool ships with, would have overstated precision by nearly half. The safer estimator is pre-selected and the naive one remains available beside it, so the user can see the difference rather than be protected from it.

VERIFIABILITY

True values shown

Demo data has known coefficients, printed under the table. The user can audit the estimator, not just consume it.

REPRODUCIBILITY

Seed exposed

A visible seed field makes any result re-creatable. Hiding randomness makes a tool feel magic; exposing it makes it scientific.

DIAGNOSTICS

Beside, not below

VIF and Breusch-Pagan sit next to the coefficients, so the assumption checks are read with the estimates, not after them.

Warnings are content, not decoration

Every fit returns a warnings list rendered as amber panels: rank deficiency, ill-conditioning, complete separation, class imbalance, overdispersion, excess zeros, non-convergence. They appear inline beneath the result rather than as dismissible toasts, because a toast is gone before the user has read the table it applies to.

5 · SCREEN THREE — ESTIMATOR COMPARISON

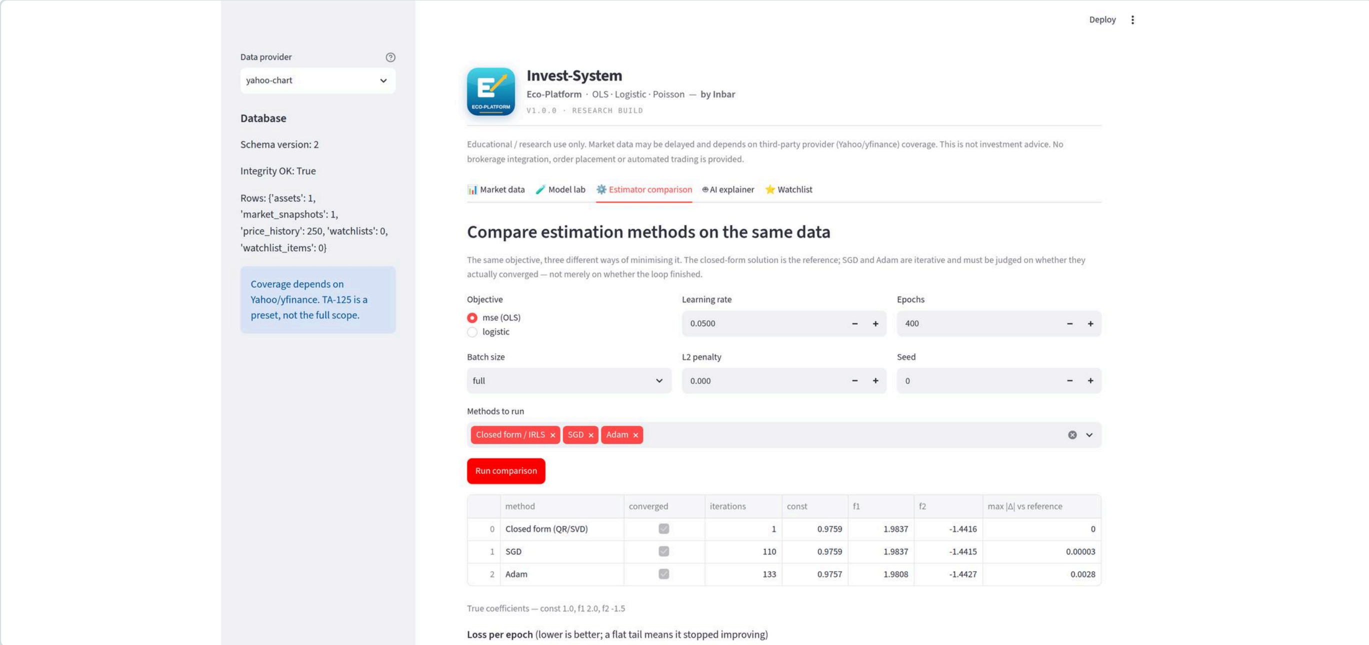


FIGURE 3 — closed-form solution against SGD and Adam on the same objective, with per-epoch loss curves and coefficient deltas from the reference.

THE SCREEN THAT EXISTS TO TEACH ONE LESSON

This tab answers a question most tools never raise: does the answer depend on how it was computed? The user ticks which methods to run, sets learning rate, epochs, batch size, L2 penalty and seed, and gets all three side by side against a trusted reference — stable least squares for OLS, IRLS for the GLMs.

"Converged" is a column, and it can say no

A finished loop is not a converged fit. The table reports convergence status, iteration count and the maximum coefficient deviation from the reference. A method that ran its full epoch budget without meeting tolerance is marked `converged = False`, and a standing note states that its coefficients should not be interpreted regardless of how the loss curve looks. Loss decreasing is not the same as loss being minimised.

CONTROLS

Every hyperparameter visible

Learning rate, epochs, batch size, regularisation, seed. Nothing is hidden behind "advanced" — the point of the screen is that these change the answer.

COMPARISON

Against a reference, always

Iterative results are meaningless in isolation. The closed-form column is the yardstick, and the delta against it is a first-class output.

Multi-select rather than a single choice

Methods are chosen with checkboxes, not a dropdown, because the screen's value is simultaneous comparison. A dropdown would frame them as alternatives to pick between; checkboxes frame them as things to run together and contrast — which is the actual task.

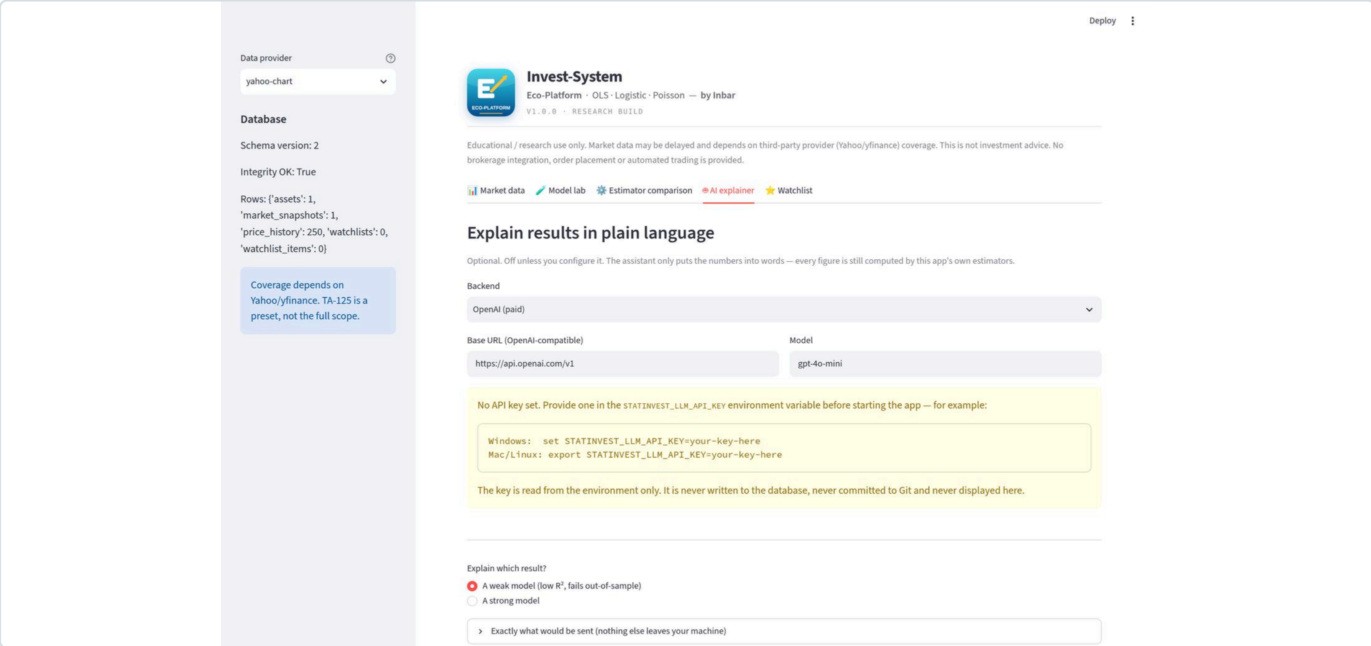


FIGURE 4 — the assistant with no key configured. The disabled state is informative rather than merely blocked.

This is the only feature that sends anything off the machine, so its interface is built around consent rather than convenience.

Show the payload before it is sent

An expander labelled *"Exactly what would be sent"* renders the literal JSON. Underneath, a note explains that this is an allow-list rather than a filter: a field reaches the API only if it is explicitly named, so internal fields cannot leak by default. The user consents to specific content, not to a vague capability.

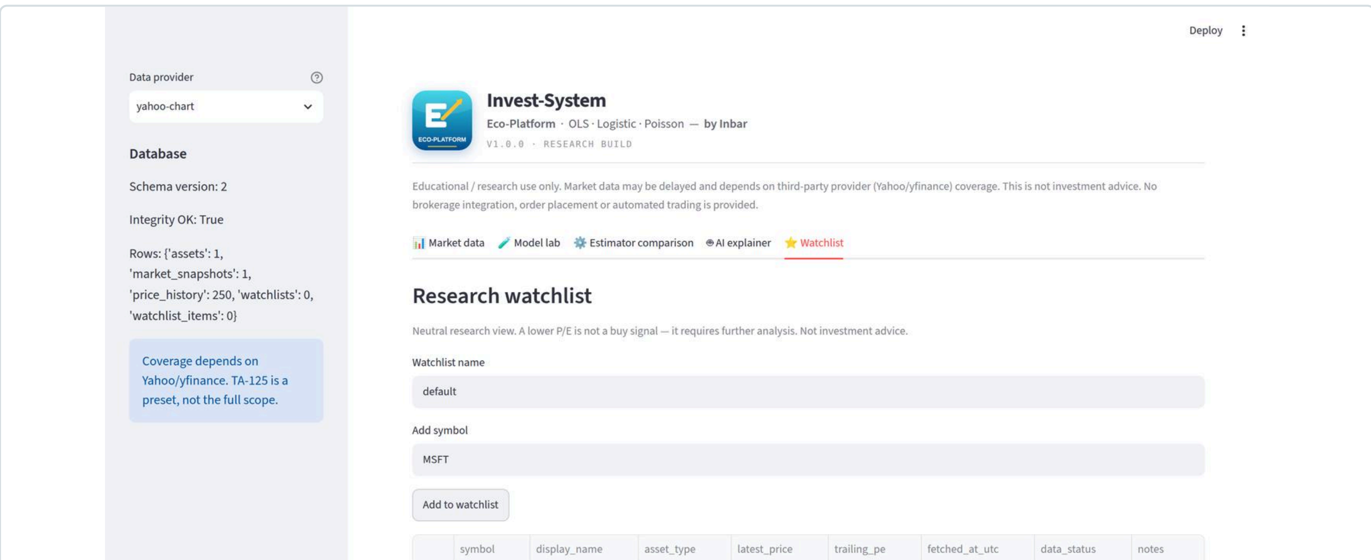
The disabled state teaches instead of blocking

With no key configured the button is disabled — and the panel shows the exact command for both Windows and Mac, states the environment variable name, and confirms that the key is never written to the database, committed, or displayed. A greyed button with no explanation is a dead end; this one is a set of instructions.

The free path is signposted, not buried

A backend selector lists local options (Ollama, LM Studio) alongside paid ones, and a standing note explains that switching to a local model costs nothing and stops data leaving the machine entirely. Presenting the cheaper, more private option as a first-class choice is a deliberate stance — the interface should not quietly steer toward the option that bills the user.

7 · SCREEN FIVE — WATCHLIST



8 · VISUAL SYSTEM

Palette

- Azure #1EC8FF — brand, logo ground
- Deep azure #004EA8 — gradient base
- Gold #FFC12E — the single accent
- Green — holds up / passed
- Red — collapses / failed
- Amber — warning, caveat

Type

Three roles, each earned. **System sans** for structure and labels. **Serif** for explanatory prose, which signals "read this" rather than "scan this". **Monospace with tabular figures** for every number, so digits align in columns and a coefficient table can be read down as well as across.

Semantic colour is reserved

Green, red and amber mean pass, fail and caution only. They are never used to distinguish one data series from another — that would make a chart legend and a verdict indistinguishable at a glance.

9 · THE INTERFACE AS A STATISTICAL SAFEGUARD

Collected in one place, these are the specific ways the UI prevents a plausible-looking mistake. Each one is a design decision that costs the user a little speed.

THE MISTAKE	HOW THE INTERFACE BLOCKS IT
Trusting an in-sample score	Out-of-sample and baseline columns sit beside the in-sample one; a verdict states plainly whether it holds or collapses.
Believing a non-converged fit	Convergence is a visible column that can read no, plus a standing warning against interpreting such coefficients.
Overstating precision	HC3 robust errors are the default; the classical option stays visible for comparison.
Mistaking demo data for real	A permanent provenance badge, with SYNTHETIC as its own labelled state.
Assuming a missing P/E is a bug	An explanatory note naming the asset type and why the ratio is undefined for it.
Reading a screen as advice	Disclaimer under the header on every screen; neutral copy; no ranking or signal styling anywhere.
Unverified currency conversion	Raw value, normalized value and the applied rule are all retained and shown in the audit expander.

10 · HONEST GAPS

ACCESSIBILITY

Not yet audited

Colour is always paired with text, but no formal contrast audit or screen-reader pass has been run. This is the first thing to fix before any public launch.

RESPONSIVE LAYOUT

Desktop-first

The layout targets a wide viewport. Narrow windows reflow via the framework's defaults rather than a designed mobile breakpoint.

MODEL LAB DATA SOURCE

Synthetic only

The lab fits generated data with known parameters — excellent for verifying the estimator, but not yet wired to the fetched market series.

CHARTING

Minimal

Results are tabular. Residual plots, leverage plots and coefficient intervals would be read faster as charts than as numbers.

The one change that would most improve the experience

Connecting the Model lab to the Market data tab. Today a user fetches a real security, then fits a model on synthetic data — two halves of a workflow that do not yet meet. Closing that gap would turn five good screens into one coherent product, and it is a larger UX win than any amount of visual refinement.